Prof. Frederic T. Chong
Department of Computer Science
University of Chicago
5730 S. Ellis Ave
Chicago, IL 60637

February 20, 2026

Dear Prof. Chong,

I am writing to propose a consulting engagement with your research group, where I can contribute hands-on AI and ML systems engineering to accelerate your work at the intersection of quantum computing and practical AI infrastructure.

Your group's focus on bridging quantum and classical computing architectures, from the EPiQC Expedition through Infleqtion's quantum software stack, aligns directly with my experience building and operating ML systems on leadership-class computing infrastructure. At the Texas Advanced Computing Center, I spent five years orchestrating distributed workloads across 800,000-node supercomputer clusters, managing multi-million-dollar compute budgets, and co-teaching graduate courses in ML, CUDA, and C++ – the exact stack that matters when pushing novel computing paradigms toward practical scale.

Since then, I have shipped production ML systems end-to-end at Homecastr (deep generative models on noisy tabular data) and conducted research at Columbia on latent factor models with Dir. Financial Engineering Ali Hirsa and on DeepSeek's GRPO algorithm with Dir. Electrical Engineering Zoran Kostic. Every engagement has required bridging the gap between research ideas and reliable, scalable systems – the same challenge your group faces in making quantum-classical hybrid architectures practical.

I would welcome the opportunity to discuss how I can contribute to your group's current projects, whether in ML infrastructure, compiler-level optimization, or systems engineering for hybrid quantum-classical workloads.

Sincerely,

Daniel Hardesty Lewis

Attached: curriculum vitæ

Daniel Hardesty Lewis
500 Riverside Dr – New York, NY 10027 – United States
+1 (713) 371-7875 • *daniel@homecastr.com* • *dlewis.ai* • *dhardestylewis*
dhardestylewis • *Google Scholar*

Daniel Hardesty Lewis

AI Consultant – ML systems, HPC, & scientific computing

AI consultant specializing in ML systems engineering, HPC infrastructure, and bridging research to production. Hands-on experience shipping deep generative models, orchestrating distributed workloads at supercomputer scale, and teaching ML, CUDA, and C++ at the graduate level.

Core Expertise

8 years **ML Systems Engineering**: Build and operate ML training and inference pipelines, including deep generative models, rigorous evaluation, and automated retraining

8 years **HPC & Distributed Computing**: Scale batch workloads across leadership-class clusters, profile bottlenecks, and optimize latency & throughput with explicit cost-performance tradeoffs

8 years **Backend & Data**: Build schemas and ETL; ship API-backed services using Python, SQL, and Postgres/PostGIS

10 years **Scientific Computing**: Spatiotemporal modeling, numerical methods, and simulation-based decision-support across geospatial, financial, and climate domains

8 years **Technical Leadership**: Drive cross-functional execution via specifications and governance; mentor engineers and researchers through design reviews, code reviews, and instruction

Technologies

Modeling	PyTorch; TensorFlow; scikit-learn; Transformers; LoRA/QLoRA/PEFT
LLM apps	embeddings; RAG; LangChain; LangGraph; DSPy; FAISS
Data stack	Python; SQL; Pandas; Polars; PyArrow; Parquet; Postgres; PostGIS; Redis; Elasticsearch; Supabase
Infrastructure	Linux/Unix; Bash; Git; Docker; Kubernetes; Modal; GCS; Ray; Spark; Dask; Airflow; CI/CD; observability
Systems	CUDA; C++; MPI; OpenMP; SLURM; containerized HPC

Experience

Professional

Nov. 2024 – Present **Co-founder**, *Homecastr*, New York, NY

Consumer real-estate pricing and forecasting platform with interactive uncertainty fancharts and temporal leakage-safe expanding-window evaluation.

- Shipped a nationwide, tract-level forecasting system, achieving an 8% one-year forecast error that beats Zillow's 8.4% industry benchmark, and proving long-term stability with a 25% median error at a 4-year horizon (a timescale competitors don't offer).
- Built an advanced diffusion model from the ground up, designing custom architecture to ensure realistic price trajectories and reliable confidence intervals across the United States.
- Created a custom evaluation framework to validate model confidence and reliability (coverage, sharpness, and PIT histograms) prior to user delivery.
- Owned the complete technical workflow—from data engineering to the live web UI (homecastr.com)—while refining go-to-market strategy based on feedback from top proptech investors including a16z and MetaProp.

Nov. 2024 – Present **Co-founder**, *PoliBOM*, Seattle, WA

AI tariff mitigation and trade compliance product for manufacturers, centered on BOM analysis, retrieval, and policy scenario simulation

- Specified an agentic workflow and schema definitions for BOM parsing, retrieval, and tariff simulation using Airflow, Elasticsearch and embeddings, and FastAPI serving backed by Supabase and Redis.
- Directed development of a **conversational AI interface** simplifying global trade workflows; enabled decision-makers to instantly **retrieve BOMs and receive proactive compliance alerts**.
- Executing customer discovery with the founder of ScoutyAI and tariff economists at Columbia Business School to validate product-market fit and pricing assumptions.
- Secured market validation by presenting the prototype to KPMG, pre-dating their subsequent launch of a commercial tariff-modeling product.

500 Riverside Dr – New York, NY 10027 – United States
+1 (713) 371-7875 • daniel@homecastr.com • dlewis.ai • dhardestylewis
dhardestylewis • Google Scholar

Nov. 2023 – Present **Founder**, *Summit Geospatial*, New York, NY
High-resolution elevation data and web delivery for engineering and hazard workflows.
- Engineered a statewide seamless Texas DEM mosaic, resampling 0.5 m LiDAR sources to 1.2 m via nearest-neighbor across 70+ state, federal, and local elevation datasets.
- Developed the web distribution platform end-to-end including data pipeline, tiling, hosting, delivery UX to support engineering and planning use cases.
- Commercial discussions to license elevation datasets to research and AI organizations.

Aug. 2021 – Aug. 2023 **Senior Data Scientist & Technical Lead**, *Texas Advanced Computing Center (TACC)*, Austin, TX
- Co-founded summitgeospatial.com to commercialize and sustain state-level terrain models developed for the $40M Texas Disaster Information System (TDIS), delivering precision decision-support for civil engineering.
- Developed inter-agency RFP and MOU frameworks with state and federal partners, translating technical requirements into scope and milestones facilitating Real-Time Inundation Mapping and TDIS.
- Scaled climate and flood models on supercomputers, orchestrating 800,000-node distributed jobs across containerized HPC clusters while managing multi-million-dollar compute budgets.
- Developed efficient methods to produce high-resolution 1m flood maps from National Water Model outputs for Texas Emergency Management's Real-Time Inundation Mapping.
- Partnered with the US Army Corps of Engineers to statistically model compound flood hazards in coastal Texas, integrating high-resolution topographic data with hydrological models.
- Contributed fundamental research supporting Paola Passalacqua's 2022 EGU Bagnold Medal, one of geomorphology's highest honors.
- Served as Technical Reviewer for Frontera Doctoral Fellows (2023) and Large-Scale Computing Pathways (2022) applications, evaluating $10M+ in computing resource allocations on one of the world's largest supercomputers.

Feb. 2018 – Jul. 2021 **Data Scientist & Research Engineer**, *Texas Advanced Computing Center (TACC)*, Austin, TX
- Competed in DARPA World Modelers to improve disaster resiliency and food security modeling in East Africa.
- Mentored Petrobras engineers in deep learning and HPC to enable institutional technology transfer.
- Received research recognition from AAAI President Yolanda Gil during her farewell address for contributions to automated scientific modeling.

Research

Jan. 2025 – Oct. 2025 **Research assistant to Dir. Financial Engineering Ali Hirsa (Industry-sponsored research)**, *Columbia University*, New York, NY
- Developed a multi-asset CVAE latent factor model with Skew-T Mixture priors; achieved 85% R^2 vs. commercial SaaS (75%) and Fama-French (58%) on backtested holdout.
- Derived a tractable, axiom-compliant per-feature SHAP algorithm using Hessian-based recursive clustering to provide stable temporal regime-dependent attributions for deep learning models.

May 2025 – Aug. 2025 **Research assistant to Dir. Electrical Engineering Zoran Kostic**, *Columbia University*, New York, NY
- Math benchmarking and finetuning of Deepseek's GRPO algorithm

July 2019 – Aug. 2019 **Summer research intern to Pres. AAAI Yolanda Gil**, *University of Southern California*, Los Angeles, CA
- Integrated hydrological models into the MINT Platform during a year-long competition with MITRE to provide DARPA the "Airflow for science"

Teaching

Jan. 2018 – Dec. 2020 **Co-instructor**, *The University of Texas at Austin*, Austin, TX
- Helped design and teach graduate-level courses: *Machine Learning for the Geosciences* for Petrobras engineers, *Intelligent Systems for the Geosciences*, and *Scientific Computation*, including C++, CUDA, and HPC.

Select Publications

2021 Gil, Y., et al. (incl. **D. Hardesty Lewis**), "Artificial Intelligence for Modeling Complex Systems: Taming the Complexity of Expert Models to Improve Decision Making". *ACM TiiS*, 2021. DOI: 10.1145/3453172

2019 Garijo, D., et al. (incl. **D. Hardesty Lewis**), "An intelligent interface for integrating climate, hydrology, agriculture, and socioeconomic models". *ACM IUI '19*, 2019. DOI: 10.1145/3308557.3308711

Education

expected 2026 **M.S., Urban Planning**, *Columbia University*, New York City, NY

2017 **B.S., Pure Mathematics**, *University of Texas*, Austin, TX

2017 **Certificate, Scientific Computation**, *University of Texas*, Austin, TX

Key Projects

oppsAlert Developed the first property- and owner-level ML model of NIMBYism, using 15+ years of 15,000+ points of individual homeowner protest data provided by the City of Austin.

500 Riverside Dr – New York, NY 10027 – United States
+1 (713) 371-7875 • *daniel@homecastr.com* • *dlewis.ai* • *dhardestylewis*
dhardestylewis • *Google Scholar*

Activities

Sailing Volunteer Atlantic SAR of the Cheeki Rafiki

Bikepacking US West & East Coasts

Languages

Spanish Native Proficiency

French Professional Working Proficiency

500 Riverside Dr – New York, NY 10027 – United States
+1 (713) 371-7875 • *daniel@homecastr.com* • *dlewis.ai* • *dhardestylewis*
dhardestylewis • *Google Scholar*